

Interaction with Immersive Cultural Heritage Environments: Using XR Technologies to represent multiple perspectives on Serralves Museum

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ABSTRACT

Museums have increasingly been using digital approaches to explore new ways to provide new experiences with Cultural Heritage (CH). The need for these solutions exploded with the COVID-19 pandemic forcing museums and cultural organizations to move towards a digital transformation to engage their audiences. Although, with a lack of guidelines on how to create eXtended Reality (XR) experiences with multiple perspectives for CH environments. This project aims to provide Museums with novel approaches to include interactive and immersive activities targeted at the cultural assets of their exhibitions and educational activities. Firstly, we will map and critically access current participatory practices in museums; then, we will develop a new methodology for creating and implementing experiences using XR technology in a range of workshops. The concept of multiple perspectives on cultural heritage will be explored through the application of co-creation practices, towards engaging different communities with cultural heritage contents from Serralves Museum.

CCS CONCEPTS

•Applied computing~Arts and humanities~Media arts • Computing methodologie~Computer graphics~Graphics systems and interfaces~Mixed / augmented reality • Computing methodologies~Computer graphics~Graphics systems and interfaces~Virtual reality

KEYWORDS

Cultural Heritage, eXtended Reality, Immersive Environments, Serralves Museum, Multiple Perspectives

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1 STATE OF THE ART

The Museums cultural environments, in the last years, have been promoting new, more engaging immersive and multi-sensorial approaches [1], to permit a direct way to the information via playful and exciting dimensions, prevail over the traditional hierarchy on exposition space, contents and visitors [2, 3].

Multiple perspectives create the possibility to help reveal context and process underpinning cultural artefacts [4], encourage exploration in the exhibition space [5] and experience personalized interactive stories [6].

A multimedia approach to the diffusion, communication, and exploitation of CH is a well-established, worldwide trend, that will grow in a post-COVID19 scenario. Several studies demonstrate how the use of new and combined media enhances the way culture is experienced [7]. These benefits the number of people (including disabled persons) who can have access to knowledge, the quality of the diffusion and the knowledge experience itself. Accordingly, XR [8] is the most used in the CH field.

The XR enabling technologies range from audio-visual display technologies, tracking and registration systems, interaction methods, and virtual environment modelling techniques [9, 10]. XR technologies and interaction methods are being used for educational, explorative, and exhibition enhancement purposes [11, 12].

Improving cultural knowledge in XR CH applications requires that the immersive reality technology and interaction method permit a contextual connection, collaboration, and engagement between users and the virtual environment [13, 14].

In The Museum of Pure Form [15], a system for visitors allows them to interact with digital models of some sculptures through touching sense; The Leventis Museum uses an application that gives the user the chance to see the virtual replicas of archaeological artefacts superimposed on a surface, while they can use their hands to interact with them in real time [16]. In the “Museo Virtuale della Via Flaminia Antica”, a VR system is used to allow visitors to move through Livia’s Villa, in its current archaeological conditions and the previous ones [17]. In Portugal, companies like byAR[18] are starting to develop CH Augmented Reality (AR) apps, and ARs apps are becoming available in some

of the Portuguese CH landmarks, like Quinta da Regaleira, which provides access to digital content (animations, 3D avatars) and narratives [19].

All the mentioned projects or experiences are focused mainly on a single perspective, lacking a multiple perspective approach. With the use of multiple perspectives, we aim to provide the visitor with different points of view of the artworks, such as the artist's point of view and the curator's point of view.

Serralves Museum in Porto which will be used as a case study in our project leads a multidisciplinary approach in contemporary art. Serralves Museum aims to create multiple relationships with art forms, nature and with all the people [20]. Serralves OnLine Experience was created in 2020 [21]. Serralves Collection Management treats, studies, documents and sort pieces or collections that, by loan, deposit, donation or even acquisition, are incorporated in the Museum and maybe in the Exhibitions or storage. Most of these artworks haven't been exhibited. [22]. Museums need to obtain solutions that boost their digital presence, transform their day-to-day functions and processes, explore with 3D digitization and AR/VR applications for visitors, exploit the creative potential of technology to offer transformative experiences for audiences or test new business models for their post-crisis recovery[23].

2 OBJECTIVES

The present study will identify elements to allow Museums to plan and structure interactive and immersive experiences representing multiple perspectives of artworks or other museums collections. Several questions arose: How to structure communication actions for creating, supporting, and maintaining digital environments for musicological institutions? What are the problems of museum interaction and communication found in digital museum environments? Therefore, the general objective of this research will be to propose a strategic theoretical plan for museum communication in immersive and interactive digital environments of museums, using XR technology, from the resources and specific characteristics of Information Design for the creation, production, and maintenance of museum object experiences in the context of interactive digital environments.

The general objective is divided into the following specific research objectives (RO):

RO1

Survey the basic principles of the history and epistemology of Design.

Study the methodology of User-Centred Design (UCD) and its delimitations.

Understand the dynamics of access to information, communication, and interaction in an XR system.

Present examples through exploratory and descriptive studies of how digital technologies create interactive, inclusive and engaging experiences in musicological and cultural institutions.

RO2

Point out the problems of museological communication in digital environments.

Learn about the use of digital technology by museum visitors, the current use of immersive and interactive systems and perceptions about the use of them and identify limitations and advantages these systems bring to visitors.

RO3

Implement experiences, using XR technology in the Serralves Museum.

Evaluate a select number of digital immersive and interactive museum experiences using the UCD.

RO4

Developing a framework to assist the creators with the design and development of XR experiences with multiple representations.

RO5

Contribute to disseminate knowledge and results by presenting workshops to professionals of museums and creators on how to create XR experiences and test our framework.

3 METHODOLOGY AND APPLIED RESEARCH METHODS

The research strategy adopted to answer the research questions is based on three components: a literature review (RO1), empirical study on the use of XR technology in visits to museums (RO2), design and implement the XR prototype experiment (RO3) and development of a framework (RO4). Each of these components is described in the following paragraphs.

RO1

The research questions presented directed the literature review to two distinct components. On the one hand (RO1.1), it is essential to understand the type of digital experiences carried out in museums. This implies understanding the aspects of digital museums, the concept of experience and how to represent multiple perspectives, namely the trends of evolution of these aspects. On the other hand, this research focused on the development of experiences to support visits to museums that would appeal to the participation of its users. Thus, we will perform a literature review to identify and compare the existing immersive reality technologies (video and sound capture, tracking and registration, audio-visual display technology, interaction methods, and virtual modelling environment methods) and identify and analyze existing interactive experience in immersive museums environments. To be able to assess experiments (RO1.2), we will review the theory of experience design and interactive design.

RO2

To identify and understand the use of digital technology in experiences in visits to museums we will use questionnaires and interviews. The questionnaire (RO2.1) will be used to assess, with visitors and museum staff, the level of use of digital technology, to identify and characterize which digital services have already been used and to discover the receptivity to participate in experiences during a visit to a museum. The interviews (RO2.2), of an exploratory nature, will be carried out to reveal different perspectives on the object of study and of helping to fine-tune the research problem and the completeness of the analysis model. Professionals and researchers in the field of museology will be

interviewed to know their perspectives on the relationship between museums, visitors and XR technology. At the same time, we will collect and analyze XR experiences that work remotely online and are designed for museum CH content.

RO3

In this objective, we will select, design (RO3.1), implement (RO3.2) and assess interactive experiments in immersive environments (RO3.3), using XR technologies, to represent multiple perspectives of CH: The prototype experiments will take place at the Contemporary Art Serralves Museum.

Having worked with the Serralves Museum in two previous projects [24,25], we have already identified the opportunity of developing and delivering multiple perspectives, using XR technologies, on the CH objects from the Museum collection that is stored. Thus, artworks in storage will be selected with the help of the Serralves Museum curators, according to their feasibility to becoming XR experiences, by checking their content and the artist's availability (RO3.1.1).

The experiments will be implemented based on the User-Centred Design (UCD) methodology, in which potential end-users of the product participate in the various phases of the development cycle [26]. The activities that make up this methodology are defined in the international standard ISO 13407: human-centred design process [27]. The main activities are: specify the context of use, specify requirements, produce design solutions, and evaluate designs. The product development cycle is iterative and is finished when all the activities mentioned above are completed. Implementation experiments will be evaluated by visitors to the Serralves Museum, using questionnaires, to assess their use in a context very close to a real usage scenario. The use of multiple perspectives will allow the visitors to observe the context and process of the artworks, experience interactive stories and environments and encourage exploration in the exhibition space. We will digitalize content from the artworks (RO3.1.2), texts, pictures, audio, video and 3D photogrammetry (if possible) to include in XR experiences prototypes (RO3.2.1). These prototypes will be tested in the Serralves Museum to conditions of light, space and others (RO3.2.2) so that we can adjust before creating the final version of the experiences (RO3.2.3).

The final version of the prototypes, the experiences, will be installed and exhibited with the support of Serralves Museum (RO3.3.1), in both physical and online spaces if needed to address the pandemic situation (partially and fully Immersive VR). We will test, validate and assess the dynamics of the interaction in the immersive environment and the adequacy of the characteristics of the contents and narratives developed to find out if this system could be a facilitating instrument in the access and representation of multiple perspectives during a visit to a museum. We will gather data using questionnaires from the public that will attend the exhibition, both presential or online (RO3.3.2). Digital presence allows us to minimize the effects if the Covid-19 pandemic remains present (RO3.3.3). RO4

From the experiences exhibition, we will obtain the functional and technical requirements for our framework that will be used to

create framework architecture (RO4.1), to assist museum professionals (curators and creators) in the development of future XR experiences, creating discourses that are understandable, comprehensive and that enable ordinary citizens to absorb information that they would not otherwise be able to, connected to the goals of "Action for CH" [28]. The framework will be implemented (RO4.2) and validated through interviews with creators and curators (RO4.3).

RO5

While developing this project we will also aim to carry out workshops targeted at museum professionals and creators to demonstrate our results (RO5.1), towards recreating them based on our project framework. Simultaneously, we will publish papers to disseminate the research (expected two per year) (RO5.2) and will do the writing of the PhD thesis (RO5.3).

4 RESULTS OBTAINED AND WORK IN PROGRESS

The first phase of the literature review (RO1.1) has already begun and, we created an excel table that would answer 23 questions: Which Year; Title; Duration; Country where it was developed; Place where it was tested; Technology used; Objectives of the experience; Heritage Target (Artifacts / Artworks / Monuments/ Landmark Building); Kind of experience (Indoor / Outdoor); Content Media used (3D Models / Audio / Pictures / Texts / Video); Platform where it runs (Desktop / Mobile / Proprietary); Software used to create; Hardware required to run the experience; Results obtained (Prototype / Pilot / Platform / Final Product); was evaluation made?; Methods of Evaluation used (If an evaluation was made); Application Area (Education / Entertainment / Heritage / Museology / Tourism); is gamification present?; is storytelling present?; does it have Information Visualization?; does it have IoT (Internet of Things)?; does it use AI (Artificial Intelligence)? and finally, in what kind of TRL (Technology readiness level) is. We applied this table to projects and experiences within the areas of CH museums. This process will allow identifying key collaborative and multimodal XR technologies and interaction methods that can be used in a CH application and the different kinds of users and their needs, according to the analyzed experiences. This analysis process will also be applied to the experiences that we will develop. By using this table we have collected already some results [29] that were presented at the International Conference "Arte, Museu e Culturas Digitais" Lisbon, 22 and 23 of April 2021. Later we did a systematic review of eXtended Reality (XR) Experiences in Museums for Cultural Heritage which answered three questions:

RQ1: What is the total distribution and volume by geographic source, location and time of issued studies on XR Technologies in Museums Experiences?

RQ2: What authors, journals, and research articles have had the highest impact on studies focusing on XR Technologies in Museums Experiences?

RQ3: What type of experiences are being developed with XR technologies in Museums Experiences?

These findings were presented at the International Conference on Intelligent Technologies for Interactive Entertainment (EAI Intertain 2021 - 13th) in December of 2021 and resulted in a book chapter [30] published in March 2022. At the same time, we realized questionnaires in the Serralves Museum (RO2.1) to understand the familiarity with the AR technology by the visitors of the museum and the acceptance of its introduction in the cultural space. By showing the visitors a cultural AR experience we gathered 97 answers to our questionnaire, and we are proceeding with its analyses for the time being. The first results of these analyses present positive results to our investigation were 85,5% of the participants liked to use the Augmented Reality system in a cultural experience and 80,4% of the participants would like to have access to Augmented Reality systems regularly in museums and cultural organizations. These first results are presented in Figures 1 and 2.

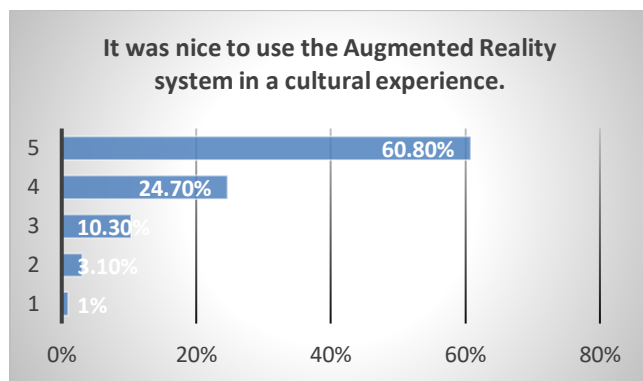


Figure 1: First analyses of the questionnaire

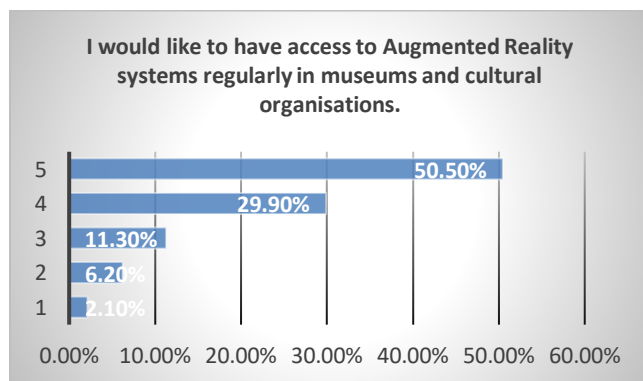


Figure 2: First analyses of the questionnaire

5 FUTURE DIRECTIONS

Future work will focus on finalizing the questionnaire analysis and completing the state of art on theory of experience design and interactive design. Next, interviews and the selection of the

artworks will be conducted with the curators of the Serralves Museum.

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